Numerical methods in Fortran

Devendra Singh Bhakuni

dbhakuni@ictp.it

/afs/ictp/public/d/dbhakuni/NM_course/NM_lec3.pdf

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Some hints for Assignment-15

1) Use transformation method to generate 10^4 random numbers from the distribution $P(x) = \frac{2}{(1+x)^2}$, $x \in [0, 1]$. Plot the histogram and compute the average $\langle x \rangle$.

- The CDF for the given function is

$$y = \int_0^x \frac{2}{(1+x')^2}\, dx' = \frac{2x}{1+x}.$$

- The inverse CDF is obtained from

$$y = \frac{2x}{1+x} \quad \Rightarrow \quad x = \frac{y}{2-y}.$$

- Get the random numbers for the desired distribution as follows:

  call random_number(V) !array with numbers between 0 and 1

  $Z = \frac{V}{(2-V)}$

- Compute the average value as: $\langle x \rangle = \mathrm{x_{avg}} = \mathrm{sum}(Z)/10^4$

Some hints for Assignment-15

2) Generates 10^6 points in the interval $[0, 1]$ from the distribution $f(x) = 3x^2$ using the transformation method and plot the histogram.

- The CDF for the given function is $y = \int_0^x f(x')dx' = x^3$.

- The inverse CDF will be $x = y^{1/3}$.

- Get the random numbers for the desired distribution as follows:

```
call random_number(V) !array with numbers between 0 and 1
```

$Z = V**(1/3)$

Some hints for Assignment-14

3) Make a subroutine implementing the Box-Muller method. Generate 10^4 random numbers from this distribution and plot the histograms for Z_1 and Z_2.

- Same as problem-2 with a different transformation function:

call random_number(U1)
call random_number(U2)

$$Z_1 = \sqrt{-2\ln(U_1)}\cos(2\pi U_2)$$

$$Z_2 = \sqrt{-2\ln(U_1)}\sin(2\pi U_2)$$

Z_1, Z_2- random numbers from standard normal dist.. ($\mathcal{N}(0,1)$).

The Gaussian distribution for any mean μ and variance σ^2 is

$$f(x,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{(x-\mu)}{\sigma})^2}$$

To generate random numbers from a Gaussian distribution, we use:

$$Z = \sigma Z_1 + \mu$$

Topics to be covered

- **Lec. 1**– Random number generators: middle square method, linear congruent method, built-in FORTRAN random number generators.

- **Lec. 2**– Non-uniform random numbers: transformation method, exponential and Gaussian random number generator

- **Lec. 3**– Rejection method and Monte-Carlo integration.

- **Lec. 4**– Random walk problem.

Monte Carlo methods

• Monte Carlo methods are a type of computational algorithm that uses random sampling (statistical methods) to solve a problem.

Applications:-

• Generating samples from a non-uniform probability distribution.

• Numerical integration.

Books:-

1. "*Monte Carlo Statistical Methods*" by C. Robert and G. Casella.

2. "*An Introduction to Computer Simulation Methods Applications to Physical Systems*" by H. Gould, J. Tobochnik, and W. Christian.

Non-uniform distribution: Rejection method

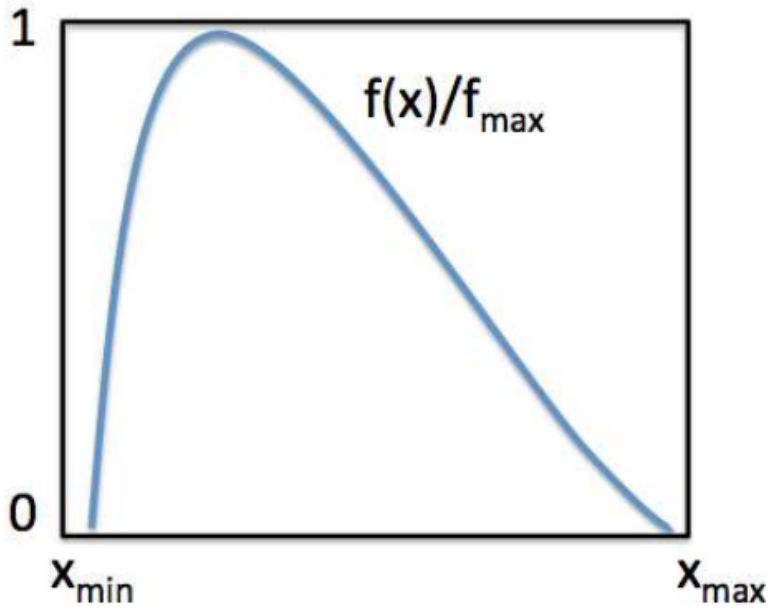
- If the inverse of the integral cannot be found.

1) generate a random number x uniformly distributed in $[x_{\text{min}}, x_{\text{max}}]$

2) generate another random number, call it r, in $[0, 1]$

3) If: $r < f(x)/f_{\text{max}}$; accept x, otherwise: reject and go to step 1.
$f_{\text{max}} = \max(f(x))$

- We need the interval and the maximum value of the PDF in this interval.



Algorithm

Data: Uniform random number x in the range $[x_{\min}, x_{\max}]$
Data: Uniform random number r in the range $[0, 1]$
Data: Target distribution with PDF $f(x)$
Result: Random number X from the target distribution

```
repeat
    Generate a random sample x from the uniform distrinution
      [x_min, x_max];
    Generate a uniform random number r in the range [0, 1];
    Calculate the acceptance probability A = f(x)/f_max;
    if r ≤ A then
        Accept X = x;
    end
until acceptance;
```

• Rejected points do not count when generating random numbers, so if you need N random numbers you need N ACCEPTED points (use do while loop)

Estimating the value of π ("Hit-or-Miss")

We can use the Monte Carlo method to estimate the value of π.

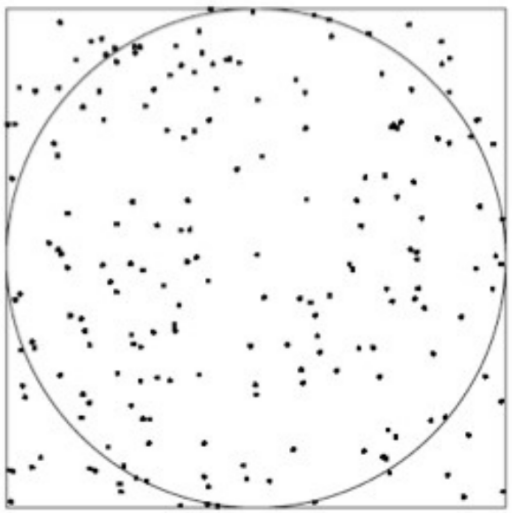
1. Generate random points (x, y) inside a square of side 2 centered at the origin $(x, y \in [-1, 1])$

2. Determine whether the point falls inside the unit circle inscribed in the square by checking whether $x^2 + y^2 <= 1$.

3. Follow step 1 and 2 for a large number of points $(N = 10^5)$

4. Calculate the ratio of the number of points that fell inside the circle (N_c) to the total number of points thrown (N_{tot}).

5. Multiply the ratio by 4 to estimate the value of π.



$$\frac{N_c}{N_{\text{tot}}} \approx \frac{A_{\text{circ}}}{A} = \frac{\pi}{4}$$

Integrating using random numbers: "Hit-or-Miss"

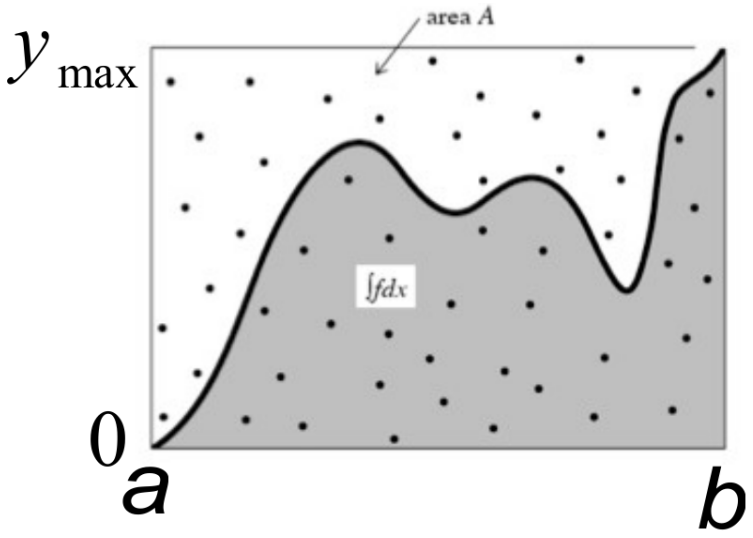
We are interested in solving the integration: $I_{\text{curve}} = \int_a^b f(x)\, dx$

- Consider a rectangular box around the curve with area: $A_{\text{box}} = y_{\text{max}}(b-a)$.

- Generate N uniformly distributed points $x_i \in [a, b]$ and $y_i \in [0, y_{\text{max}}]$.

- $x_i = a + u_1(b-a)$, $y_i = u_2 y_{\text{max}}$ (u_1 and u_2 are uniform random numbers in $[0, 1]$).

- Count N_0, the number of points which fall below the curve: $y_i < f(x_i)$.



$$\frac{A_{\text{curve}}}{A_{\text{box}}} = \frac{I_{\text{curve}}}{A_{\text{box}}} \approx \frac{N_0}{N}$$

$$I_{\text{curve}} \approx \frac{N_0}{N} y_{\text{max}}(b-a)$$

Integrating using random numbers: "Hit-or-Miss"

- We are interested in solving the integration: $I_{\text{curve}} = \int_a^b f(x)\, dx \approx A$

Algorithm

Data: Integration limits a, b, function $f(x)$, number of samples N
Result: Approximate integral value A

```
N_0 = 0;
for i = 1 to N do
    Generate a random number x between a and b;
    Generate a random number y between 0 and M, where M is
     an upper bound of f(x);
    if y < f(x) then
        N_0 = N_0 + 1;
    end
end
A = (N_0 / N) · (b − a) · M;
return A;
```

Integration using "Crude Monte-Carlo method"

- From the lectures on integration methods (for e.g., Left Riemann)

$$I = \int_0^1 f(x)\ dx \approx \sum_{i=1}^{N} f(x_i) \cdot h$$

Here $h = 1/N$, $x_i = a + (i-1)h$. **Note:** x_i **are deterministic!**

- Let's use random numbers to solve this integration. The average of a function $f(x)$ for a probability density function $p(x)$ is

$$\langle f \rangle = \int p(x)\ f(x)\ dx = \frac{1}{N} \sum_{i=1}^{N} p(x_i) f(x_i)$$

For a uniform PDF, i.e. $p(x) = 1$ for $x \in [0, 1]$,

$$\langle f \rangle = \frac{1}{N} \sum_{i=1}^{N} f(x_i) \implies \langle f \rangle = \int_0^1 f(x)\ dx$$

x_i**'s are random numbers from uniform distribution!**

Integration using "Crude Monte-Carlo method"

$$I = \int_a^b f(x)\, dx = \frac{1}{N} \sum_{i=1}^{N} f(x_i) \cdot (b-a)$$

Algorithm

Data: Number of samples N, function $f(x)$, intg. limits a and b
Result: Approximate integral value A
$A = 0$;
for $i = 1$ *to* N **do**
  Generate a random number X_i uniformly distributed in $[a, b]$;
  Calculate $f(X_i)$;
  $A = A + f(X_i)$;
end
$A = \frac{A}{N}(b-a)$;
return A;

- can also be used in higher dimensions. $X_i \to \vec{\mathbf{X}}$, and $(b-a) \to V$

Estimating Error in Crude Monte Carlo

• Let's calculate the variance σ^2:

$$\sigma_f^2 = \langle (f - \langle f \rangle)^2 \rangle = \frac{1}{N} \sum_{i=1}^{N} (f(x_i) - \langle f \rangle)^2 \, p(x_i) = \left(\langle f^2 \rangle - \langle f \rangle^2 \right)$$

"measure to which extent f deviates from its average"

• Consider the previous result for a fixed N as one measurement. Suppose we recalculate the average/variance for M different measurements. Then, we can write the integral as the average of M such averages: $I_M = \frac{1}{M} \sum_{l=1}^{M} \langle f \rangle_l$

• The variance of these measurements will be

$$\sigma_M^2 = \left(\langle I_M^2 \rangle - \langle I_M \rangle^2 \right) \qquad \text{and} \qquad \sigma_M \sim \frac{1}{\sqrt{M}}$$

This is true for any dimension d!

Estimating Error in Crude Monte Carlo

• Let's compare it to any numerical integration based on Taylor expansion. The error goes as $\mathcal{O}(h^k)$, $k = 2$ for trapezoidal rule and $h = (b - a)/N$. That means, the error goes as $\sim N^{-k}$.

• Consider integration in higher dimension: Suppose integration volume is a hypercube with side L and dimension d. The number of points in the cube: $N = (L/h)^d \implies$ the error goes as $\sim N^{-k/d}$

• If we perform the same integration using MC method, the error $\sigma \sim 1/\sqrt{N}$. Thus, for dimension $d > 2k$, MC method has better accuracy than the traditional numerical integration methods.

Assignment-16

1) Estimate the value of π using "Hit-or-Miss" with $N = 10^4$ points.

2) Generates 10^4 points in the interval $[0, 1]$ from the distribution $f(x) = 3x^2$ using the rejection method and plot the histogram.

3) Estimate the integral: $\int_0^3 3x^2 dx$ using the "Hit and Miss" method using the number of samples $N = 10^4$.

4) Estimate the integral: $\int_0^1 e^{-x^2} dx$ using "crude Monte-Carlo" method using the number of samples $N = 10^4$.

Bonus: Estimate the value of π using "crude Monte-Carlo" method by integrating the function: $F(x, y) = \begin{cases} 1 & \text{if } x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$, in the domain $\Omega = [-1, 1] \times [-1, 1]$ and with $N = 10^4$.

Hint: $\int_\Omega F(x, y) dx dy = \pi$. Generate $2N$ uniform random numbers $u, v \in [0, 1]$.

Submit your code as ***surname_assignment_16.f90*** to ***dbhakuni@ictp.it*** before the next lesson.